

Secure File Sharing System

1. Project Title

Secure File Sharing System

2. Problem Statement

Organizations and individuals need a secure method to share sensitive files over the internet. Traditional file sharing methods often lack strong authentication, encryption, access control, and activity tracking, increasing the risk of unauthorized access and data breaches.

3. Abstract

The Secure File Sharing System is a web-based application designed to securely upload, store, and share files between authorized users. The platform uses authentication, role-based access control, encrypted file storage, and secure sharing links to protect confidential data. Users can upload files, share them with selected recipients, manage permissions, and monitor download history. The application is developed using React for the frontend, FastAPI (or Express.js) for the backend, OpenAI GPT or Google Gemini for intelligent user assistance and security recommendations, and MongoDB for securely storing user information, metadata, and sharing records. The system ensures confidentiality, integrity, and controlled access to shared files.

4. Objectives

- Enable secure file uploads and sharing.
- Implement authentication and authorization.
- Protect files using encryption.
- Track downloads and sharing history.
- Improve data security and accessibility.

5. Scope

Suitable for businesses, educational institutions, healthcare organizations, legal firms, and individuals who need secure digital file exchange.

6. Existing System

Many conventional file sharing methods rely on public links or email attachments, offering limited access control and security monitoring.

7. Proposed System

The system allows authenticated users to upload encrypted files, generate secure sharing links, assign permissions, and monitor file access through an admin dashboard.

8. Modules

User Authentication, File Upload, Secure Storage, File Sharing, Access Control, Download History, Admin Dashboard.

9. Technology Stack

Frontend: React

Backend: FastAPI / Express.js

Database: MongoDB

AI: OpenAI GPT / Google Gemini

Storage: Local Storage / Cloud Storage (AWS S3 or Firebase Storage)

Security: JWT Authentication, File Encryption

10. Methodology

User Registration → Login → Upload File → Encrypt & Store → Generate Secure Link → Share with Recipient → Validate Access → Download → Log Activity

11. Expected Outcome

Provides a secure, reliable, and efficient platform for sharing confidential files while minimizing unauthorized access and improving data protection.

12. Future Enhancements

End-to-end encryption, expiring download links, multi-factor authentication, malware scanning, version control, audit reports, and mobile application support.

13. Conclusion

The Secure File Sharing System enhances digital collaboration by providing encrypted storage, controlled file access, and secure sharing mechanisms. It offers a scalable and reliable solution for protecting sensitive information in personal and professional environments.